

KeraNova Data Transformation Template

Version 1.0 for committee review on 1 October 2026

KeraNova Research Team | Version 1.0 | 1 October 2026

Purpose and analytic structure

This template transforms verified keratoconus observations into a retrospective cohort table and a model table for twelve-month CDVA prediction. It defines calendar-based endpoint selection, clinical derivations, cohort dispositions and patient-level preparation for internal validation.

Keep three linked levels: coded patient, index eye treatment episode, and dated visit or scan. One patient may contribute both eyes; every visit and image for that patient must remain in the same training or validation partition. A stable source record ID does not substitute for patient linkage.

Interval selection rules

Time point	Accepted observation	Selection rule
Baseline	Nearest valid documented preoperative value for each predictor	Preserve measurement date and lookback. Same-day requires preoperative chronology. Clinically review distant values; interval-restriction sensitivity must be prespecified.
Interim one month	Actual dated visits or original one-month source label	Descriptive only. No date-based window is inferred from CSV headers; retain actual elapsed time when retrieved.
Interim six months	Actual dated visits or original six-month source label	Descriptive only; not a substitute for missing twelve-month CDVA.
Primary twelve-month endpoint	Index date plus 10 calendar months through plus 14 calendar months, inclusive	Select valid CDVA closest in days to the 12-calendar-month anniversary. Equal distance selects earlier visit.
Other postoperative scan or safety event	Actual dated encounter	Keep separate until timing is verified. Current postop topography and haze fields have unassigned timing.

Endpoint algorithm

- Calculate the 10-, 12- and 14-calendar-month anniversaries from the verified index date. If the starting day does not exist in a target month, use that month's last day. Keep these anniversary dates in the selection log.
- Restrict candidate outcomes to the index eye, verified CDVA measurement, usable notation, and visits within the inclusive window. Preserve every candidate and exclusion reason.
- Select the candidate with the smallest absolute day distance from the 12-month anniversary. Break a tie by earlier visit date. Resolve differing measurements on the same date using documented correction and method with reviewer adjudication; never choose the better numerical result.
- Convert the selected measurement under the frozen visual-acuity rule and record source visit ID, actual date, raw notation, method, logMAR and selection reason. If no valid measurement remains, assign a missing target and keep the eye in the clinical cohort with a stated missing-outcome reason.
- Do not interpolate or carry forward labels from 1 or 6 months, UDVA, refraction, pinhole vision, another eye or a public dataset. Do not change the endpoint because of observed model performance.

Derived variables and direction

Variable	Derivation	Interpretation
cdva_12m_logmar	Verified selected endpoint; log10(d/n) for unqualified Snellen n/d	Lower values indicate better vision
cdva_baseline_logmar	Verified preoperative corrected vision with conversion rule	Primary baseline predictor
delta_cdva_logmar	cdva_12m_logmar minus cdva_baseline_logmar	Negative change indicates improvement
mrse_calculated_D	sphere_D plus cylinder_D divided by 2	Same eye and encounter; preserve reported value separately
delta_mrse_D	Selected endpoint MRSE minus baseline MRSE	Signed refractive change; not a prediction target
delta_kmax_D	Verified same endpoint visit Kmax minus baseline Kmax	Negative change indicates flattening; currently unavailable as timed endpoint
delta_tct_um	Verified same endpoint visit TCT minus baseline TCT	Record units and scan quality; thinning alone is not a declared failure
elapsed_days	Visit date minus index date	Calendar dates determine endpoint membership
elapsed_months_display	elapsed_days divided by 30.4375	Descriptive display only; do not use to replace calendar-month window
haze_present_observed	Documented positive or negative observation with date	Primary CSV haze timing is unverified; no resolution-time derivation
treatment_group	Verified TG-PRK plus accelerated CXL or accelerated CXL alone	Categorical input; associations do not establish treatment benefit

Cohort dispositions and denominators

Assign core eligibility independently from analysis availability. The screened ledger retains all eye records, all criteria, pending verification and exclusions. Use mutually exclusive primary disposition reasons for the flow diagram and retain all secondary reasons for audit. Report patient counts only after coded linkage is recovered.

The clinical descriptive cohort consists of verified adult eligible eyes receiving a qualifying treatment in the approved period. The outcome cohort contains those with a valid primary-window CDVA. The change-from-baseline subset additionally requires valid baseline CDVA. The prediction cohort requires verified patient ID and outcome plus the predictor data permitted by the frozen missingness plan. Report every denominator separately.

The proposal's 30 eyes per group is a planned maximum or target, not an observed result. The present source contains 27 PRK-with-CXL labels, 51 CXL labels and one PTK-with-CXL label before eligibility checks. Do not manufacture records, oversample to create apparent clinical observations, or select patients based on outcome values. Any restriction to a fixed sample must be prespecified with the investigator and independent of outcomes.

Observed source availability

Counts below are populated source values after excluding administrative rows; they are not counts of verified calendar-window outcomes. A hyphen or blank is missing. Qualified acuities remain populated for this audit but may be unusable for the primary numeric endpoint until reviewed.

Source interval	UDVA populated n	CDVA populated n	Verified dated endpoint n
Baseline	78	9	Not established
One-month source label	65	6	Not established
Six-month source label	62	5	Not established
Twelve-month source label	39	2	Not established

Current primary model readiness

Only CSV rows 6 and 76 contain both baseline and source-labelled twelve-month CDVA. Row 6 contains the unresolved endpoint 20/30-. Row 76 contains 20/25. Both are CXL-labelled records. Their patient linkage, correction methods, treatment protocol and visit dates are unverified. The current export provides no verified primary outcome labels and cannot estimate or validate the proposed multivariable model.

The next authorized extraction must prioritize coded patient and eye identifiers, index and visit dates, true corrected vision at baseline and within 10 to 14 months, verified treatment subtype, baseline Kmax, verified TCT and scan quality. Track recovered, unusable and missing outcomes by treatment group. Public diagnostic imaging data can support a separately labelled exploratory image demonstration, but cannot repair these missing longitudinal outcomes.

Model input and validation preparation

Freeze the candidate set to baseline CDVA, age at index treatment, sex, verified treatment group, baseline Kmax and one pachymetry metric. KeraNova prespecifies verified baseline TCT; the current thinnest loca column requires metric confirmation. Do not silently substitute CCT or apical thickness. No postoperative measurement, haze observation or follow-up availability indicator is a baseline predictor.

The primary model uses complete cases for all six prespecified predictors. An optional sensitivity analysis may impute missing baseline Kmax and TCT using training-partition medians and add two missingness indicators. Age, sex, baseline CDVA, verified treatment and patient identifiers must be observed and verified. No alternative predictor-imputation approach is added after examining validation results.

Create patient-level resampling assignments before fitted preprocessing. All data from the same patient, including fellow-eye measurements and images, stay together. Within each training partition only, learn permitted numeric imputation, categorical encoding and scaling; apply that fitted transformation to its held-out patients. Do not fit imputers or transformations once on the entire dataset.

Never impute the primary outcome. Keep raw clinical data without model imputation, and save fold-specific preprocessing parameters separately. If unknown patient IDs or insufficient verified labels prevent grouped validation, do not report internal-validation performance.

Safety events and optional secondary outcomes

Preserve dated complications and subsequent procedures through the endpoint. Do not automatically discard postoperative values after a complication; report the treatment pathway and apply the prespecified secondary sensitivity analysis when the data supports it. Haze resolution requires documented onset, dated positive and negative observations and a defined grading scale. The current CSV supports neither time-to-resolution estimates nor a validated postoperative haze grade.

Any definition of clinical progression, stability, two-line change or safety loss requires agreement on the actual chart and measurement method before analysis. LogMAR change can be reported numerically without claiming an unsupported line-equivalent threshold.

Verification and release checklist

- Reconcile administrative rows, screened eye records, unique patients, final eligibility, exclusions, pending records, available labels and each analysis denominator.
- Recompute representative Snellen conversions and MRSE independently. Check date-window boundaries, calendar end-of-month behavior and equal-distance tie selection with the actual chosen endpoint log.
- Verify all target and treatment assignments against records. Check that postoperative information is absent from the predictor matrix and that no patient overlaps training and validation partitions.
- Freeze source hash, dictionary version, cohort ledger, clean dataset, analysis specification and transformation log. Data release approvals and signatures: clinical lead _____ data custodian _____ statistician _____ date _____.

